OSS711S LAB: Student Reponse Form

1. **Cover Page**

Operating Systems Security (OSS711S) Lab Report	
Lab No:	
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Nature of Lab: (Lab Objective)	The objective of this lab was to design and implement a secure and cost-effective enterprise e-mail communication system for Virtual Connect cc. We were required to configure an e-mail server, Active Directory Domain Controller, distributed file services using Samba and NFS, DHCP services, and domain password policies while applying security best practices.
Due Date:	*Tuesday, 14 April 2026, 11:59 AM*
Description of Tools used	
VirtualBox	
Windows Server 2022	
Windows 8 Client	
Samba, NFS	
Dovecot, Roundcube	
Ubuntu Server, Postfix	
ClamAV, SpamAssassin	
DHCP Server, UFW Firewall	

2. Introduction

In this lab, we implemented different enterprise network services in a virtual environment. The main focus was to configure a secure e-mail system, shared file services, and password policies for an organization. We used Windows Server for Active Directory and Ubuntu Server for mail, Samba, NFS, and DHCP services. This lab helped us understand how different services work together in a real company environment while ensuring security and proper access control.

3. Lab Procedures

SECTION A – E-MAIL SYSTEM IMPLEMENTATION (60 MARKS)

Step 1: Creating the Virtual Machines

We first created three virtual machines in VirtualBox**:**

- Windows Server VM → Domain Controller
- Ubuntu Server VM → Mail Server
- Windows 10 VM → Client Machine

All machines were connected to the same Internal Network (LabNet)**.**

📸 **Insert Screenshot 1: VirtualBox machines created**

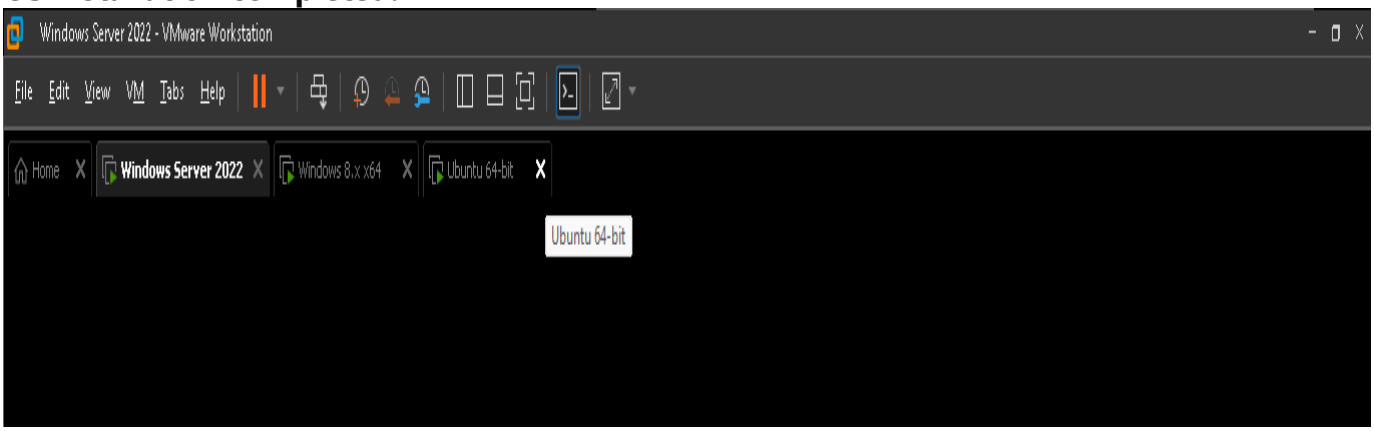
Step 2: Installing Operating Systems

We installed the following:

- Windows Server on the first VM
- Ubuntu Server on the second VM

- Windows 10 on the client VM

OS installation completed:



Step 3: Configuring Static IP Addresses

We configured static IP addresses for the servers:

IP configuration:

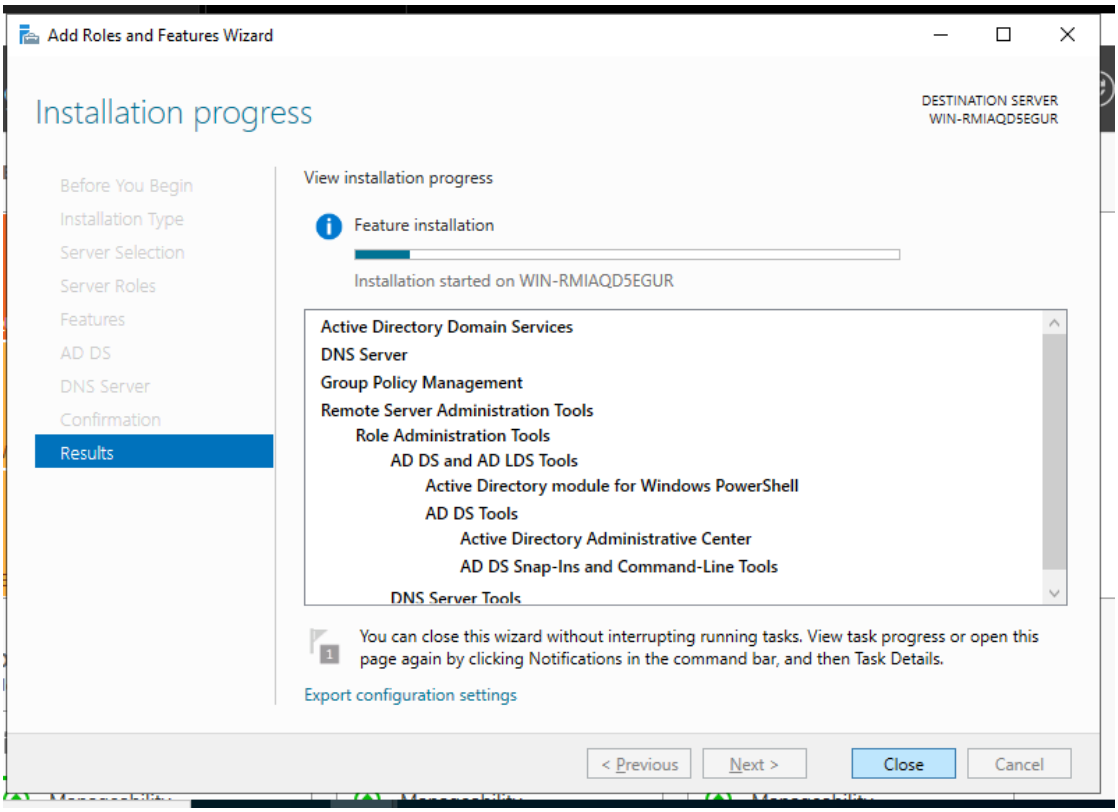
- Machine: Domain Controller — IP Address: 192.168.10.10
- Machine: Linux Server — IP Address: 192.168.10.50
- Machine: Client — IP Address: DHCP / 192.168.10.20

Step 4: Configuring Active Directory

- ✓ On Windows Server:
- ✓ Opened Server Manager
- ✓ Added Active Directory Domain Services
- ✓ Promoted server to Domain Controller

Created domain:

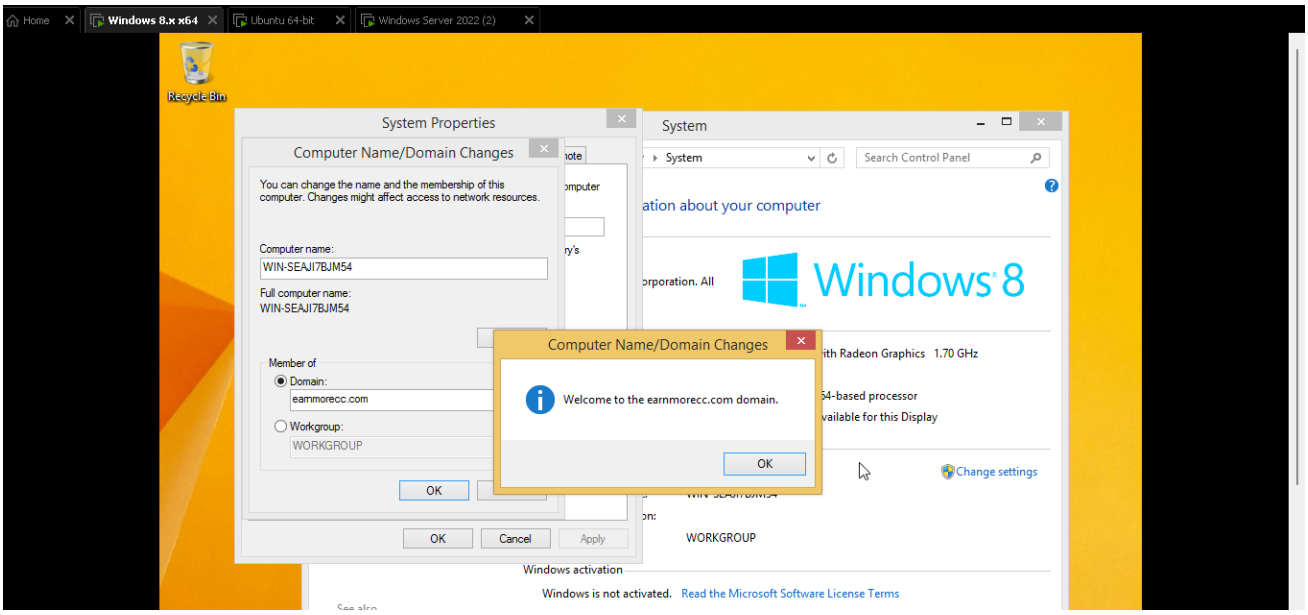
```
earnmorecc.com
```



Step 5: Joining Client Machine to Domain

The Windows client was joined to the domain:

```
earnmorecc.com
```



Step 6: Installing Mail Server

On Ubuntu Server, we installed:

sudo apt update

```
hambata@hambata-VMware-Virtual-Platform:~$ sudo apt update
[sudo] password for hambata:
Hit:1 http://na.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://na.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://na.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
70 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

sudo apt install postfix

```
hambata@hambata-VMware-Virtual-Platform:~$ sudo apt install postfix
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
postfix is already the newest version (3.8.6-1build2).
The following packages were automatically installed and are no longer req
uired:
  libconfig-inifiles-perl libdbd-mysql-perl libdbi-perl libmariadb3
  libmysqlclient21 libterm-readkey-perl mariadb-common
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 70 not upgraded.
```

sudo apt install dovecot-imapd dovecot-pop3d

```
hambata@hambata-VMware-Virtual-Platform:~$ sudo apt install dovecot-core
dovecot-imapd dovecot-pop3d
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer req
uired:
  libconfig-inifiles-perl libdbd-mysql-perl libdbi-perl libmariadb3
  libmysqlclient21 libterm-readkey-perl mariadb-common
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libexttextcat-2.0-0 libexttextcat-data
Suggested packages:
```

sudo apt install roundcube

```
hambata@hambata-VMware-Virtual-Platform:~$ sudo apt install roundcube
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
roundcube is already the newest version (1.6.6+dfsg-2ubuntu0.1).
The following packages were automatically installed and are no longer req
uired:
  libconfig-inifiles-perl libdbd-mysql-perl libdbi-perl libmariadb3
  libmysqlclient21 libterm-readkey-perl mariadb-common
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 70 not upgraded.
```

This was used for:

- ✓ Sending mail
- ✓ Receiving mail
- ✓ Web-based mail access

Step 7: Security Configuration

We installed:

```
sudo apt install clamav
sudo apt install spamassassin
sudo ufw enable
```

Allowed required ports only:

```
sudo ufw allow 25
sudo ufw allow 143
sudo ufw allow 80
```

Unused ports were closed for hardening.

Firewall and antivirus:

```
hambata@hambata-VMware-Virtual-Platform:~$ sudo ufw enable
Firewall is active and enabled on system startup
hambata@hambata-VMware-Virtual-Platform:~$ sudo ufw allow 25
Rule added
Rule added (v6)
hambata@hambata-VMware-Virtual-Platform:~$ sudo ufw allow 143
Rule added
Rule added (v6)
hambata@hambata-VMware-Virtual-Platform:~$ sudo ufw allow 80
Skipping adding existing rule
Skipping adding existing rule (v6)
hambata@hambata-VMware-Virtual-Platform:~$
```

Answers to Section A Questions

a) How many computers are required?

For this setup, we need a minimum of three computers (virtual machines). One computer is required for the Windows Server acting as the Domain Controller, one for the Linux mail server, and at least one Windows client machine for testing the system. More client machines can be added depending on the number of employees.

Minimum 3 computers:

- 1 Windows Server
- 1 Linux Server
- 1 Windows Client

b) How much mail storage space should be allocated to each employee?

A reasonable amount of storage space would be between 1 GB and 5 GB per employee. This should be enough for normal business communication, attachments, and archived messages. The exact amount can be increased depending on the company's needs and available storage capacity.

c) Which software should be used for features like recalling messages, scheduling meetings, and working offline?

The best software to use is Microsoft Outlook because it supports advanced features such as message recall, meeting scheduling, calendar management, and offline access to emails.

d) Which tools will facilitate creating, editing, or removing e-mail accounts with only a few mouse clicks?

We can use Active Directory Users and Computers (ADUC) on the Windows Server. This tool makes it easy to create, modify, disable, or delete user accounts and email accounts through a graphical interface.

e) Which server should be set up on the network?

Two servers need to be set up on the network:

- A Windows Server for Active Directory and Domain Services
- A Linux Server for the e-mail services (Postfix, Dovecot, and webmail)

f) What kind of antispam system should be installed?

An antispam system such as SpamAssassin should be installed. It helps detect and block spam, junk mail, and suspicious messages before they reach employees.

g) Do you need to install a Web-based e-mail system or a client-based e-mail system?

It is best to install both a web-based and a client-based e-mail system.
The web-based system allows employees to access emails from anywhere using a browser, while the client-based system provides more advanced features such as offline access and scheduling.

h) If a client-based e-mail system is installed, then how will the employees access the Web?

Employees will access the web through a normal web browser such as Google Chrome or Microsoft Edge on their Windows client machines. This allows them to use webmail when needed.

i) Which e-mail filtering programs can be configured to monitor e-mails and add disclaimers?

Programs such as Postfix mail filters together with SpamAssassin can be used to monitor emails, block unwanted content, and automatically add company disclaimers to outgoing messages.

j) How will you be informed of the server status and usage?

We can monitor the server status and usage by using:

- system logs
- server monitoring tools
- email alerts for administrators
- performance monitoring dashboards

These tools help us check uptime, storage usage, and service performance.

l) How will you be able to generate and view detailed HTML reports on all the e-mail activities?

HTML reports can be generated using mail log analysis tools that read the mail server logs and convert them into detailed reports. These reports can show sent emails, received emails, failed messages, and usage statistics.

m) What kind of connectivity is required between the servers and the clients?

A LAN connection using TCP/IP networking is required between the servers and client machines. In the virtual environment, all machines should be connected to the same internal network in VirtualBox.

n) Where should the mail server be deployed?

The mail server should be deployed on the Linux server inside the company network or data center. In this lab, it is deployed inside the Ubuntu virtual machine.

o) Which mail protocol should be used to search through messages while they are still on the server?

The correct protocol to use is IMAP.
This protocol allows users to search emails directly on the server and choose only the messages they want to download.

SECTION B – SAMBA AND NFS (40 MARKS)

Step 1: Creating Shared Folder

We created the required folder:

```
sudo mkdir -p /data/notes
```

📸 **Insert Screenshot 8: Folder creation**

Step 2: Creating Users

```
sudo useradd bob
sudo useradd popy
sudo useradd shuta
sudo passwd bob
sudo passwd popy
sudo passwd shuta
```

User creation

```
hambata@hambata-VMware-Virtual-Platform:~$ sudo adduser popy
info: Adding user `popy' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `popy' (1003) ...
info: Adding new user `popy' (1003) with group `popy (1003)' ...
info: Creating home directory `/home/popy' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for popy
Enter the new value, or press ENTER for the default
        Full Name []: Popy
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
info: Adding new user `popy' to supplemental / extra groups `users' ...
info: Adding user `popy' to group `users' ...
hambata@hambata-VMware-Virtual-Platform:~$ sudo adduser popy
fatal: The user `popy' already exists.
hambata@hambata-VMware-Virtual-Platform:~$ sudo adduser shuta
```

Step 3: Installing Samba

```
sudo apt install samba
```

Samba installation:

```
hambata@hambata-VMware-Virtual-Platform:~$ sudo apt install samba
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
samba is already the newest version (2:4.19.5+dfsg-4ubuntu9.4).
The following packages were automatically installed and are no longer req
uired:
  libconfig-inifiles-perl libdbd-mysql-perl libmariadb3
  libmysqlclient21 libterm-readkey-perl mariadb-common
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 67 not upgraded.
hambata@hambata-VMware-Virtual-Platform:~$
```

Step 4: Configuring Samba Share

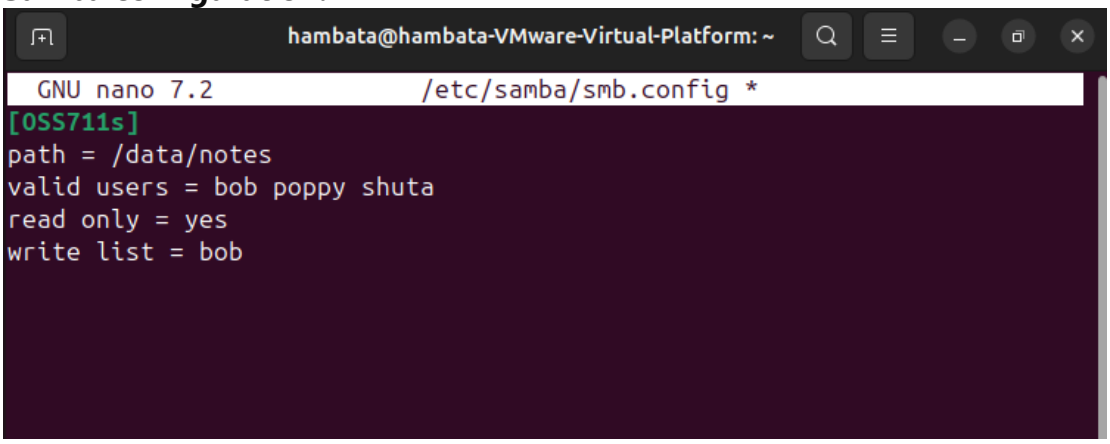
Edit:

```
sudo nano /etc/samba/smb.conf
```

Add:

```
[OSS711s]
path = /data/notes
valid users = oss admin bob
read only = yes
write list = bob
```

Samba configuration:



Step 5: Testing from Windows Client

From Windows:

```
\\192.168.10.20\OSS711s
```

Confirmed:

- bob → write access
- oss/admin → read only

```
hambata@hambata-VMware-Virtual-Platform: ~
  GNU nano 7.2                /etc/samba/smb.config *
[OSS711s]
path = /data/notes
valid users = bob poppy shuta
read only = yes
write list = bob
```

Step 6: Configuring NFS

Install:

```
sudo apt install nfs-kernel-server
```

```
hambata@hambata-VMware-Virtual-Platform:~$ sudo apt install nfs-kernel-se
rver
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nfs-kernel-server is already the newest version (1:2.6.4-3ubuntu5.1).
The following packages were automatically installed and are no longer req
uired:
  libconfig-inifiles-perl libdbd-mysql-perl libmariadb3
  libmysqlclient21 libterm-readkey-perl mariadb-common
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 67 not upgraded.
```

Create folders:

```
sudo mkdir /binary
sudo mkdir /bits
```

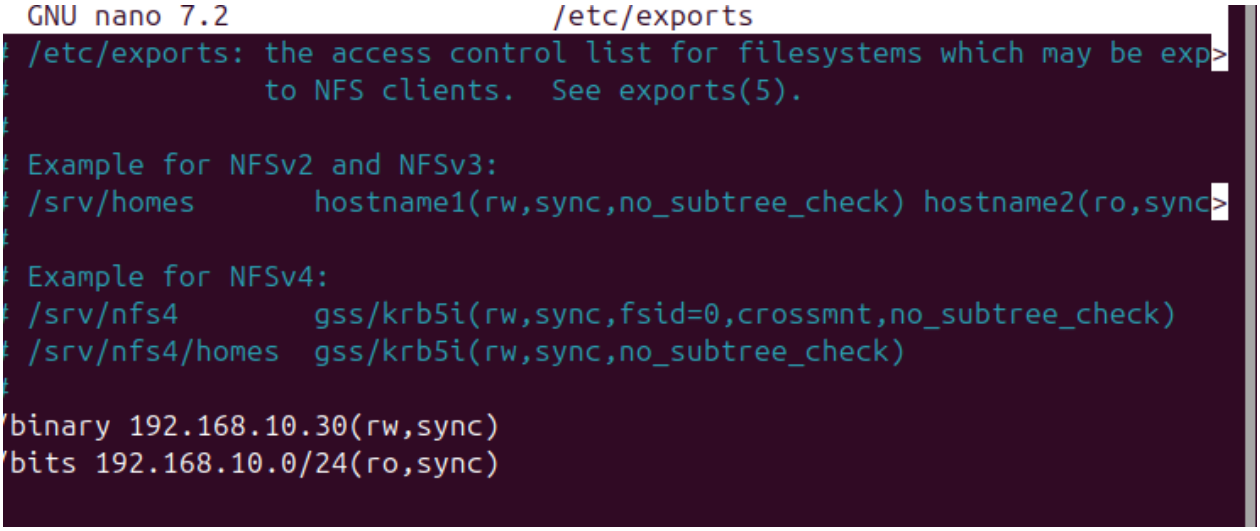
Edit:

```
sudo nano /etc/exports
```

Add:

```
/binary 192.168.10.30(rw,sync)
/bits 192.168.10.0/24(ro,sync)
```

NFS configuration



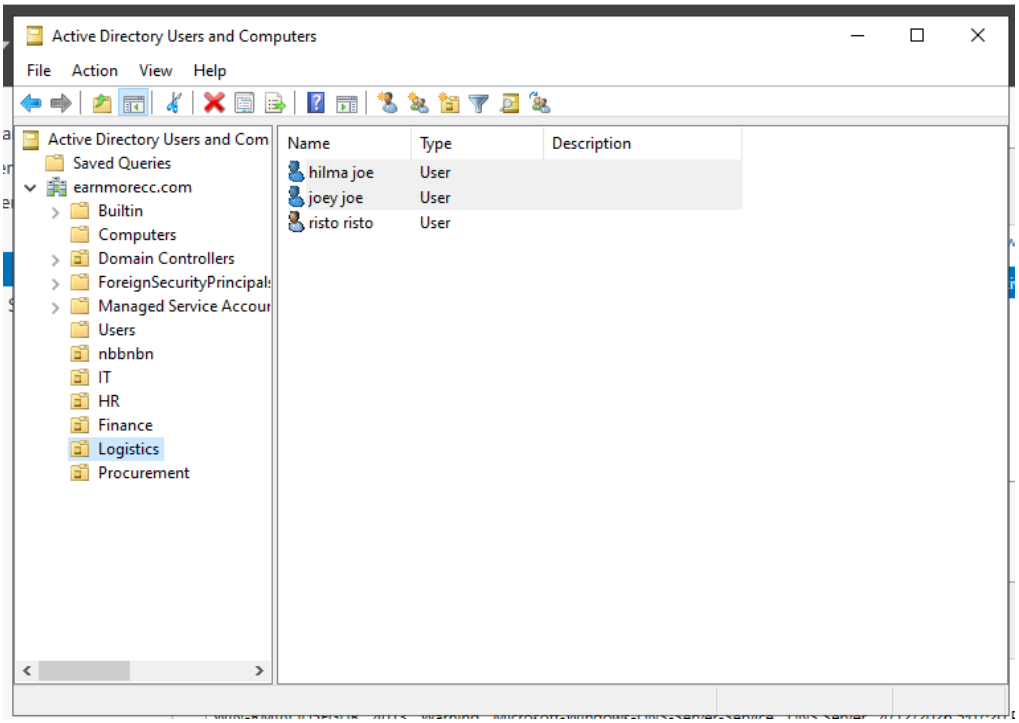
SECTION C – PASSWORD POLICY, DHCP & SECURITY (70 MARKS)

Step 1: Creating Department Users

We created:

- IT → 2 users
- HR → 2 users
- Finance → 2 users
- Procurement → 2 users
- Logistics → 3 users

Department users in AD



Step 2: Password Policy

On Domain Controller:

Go to:

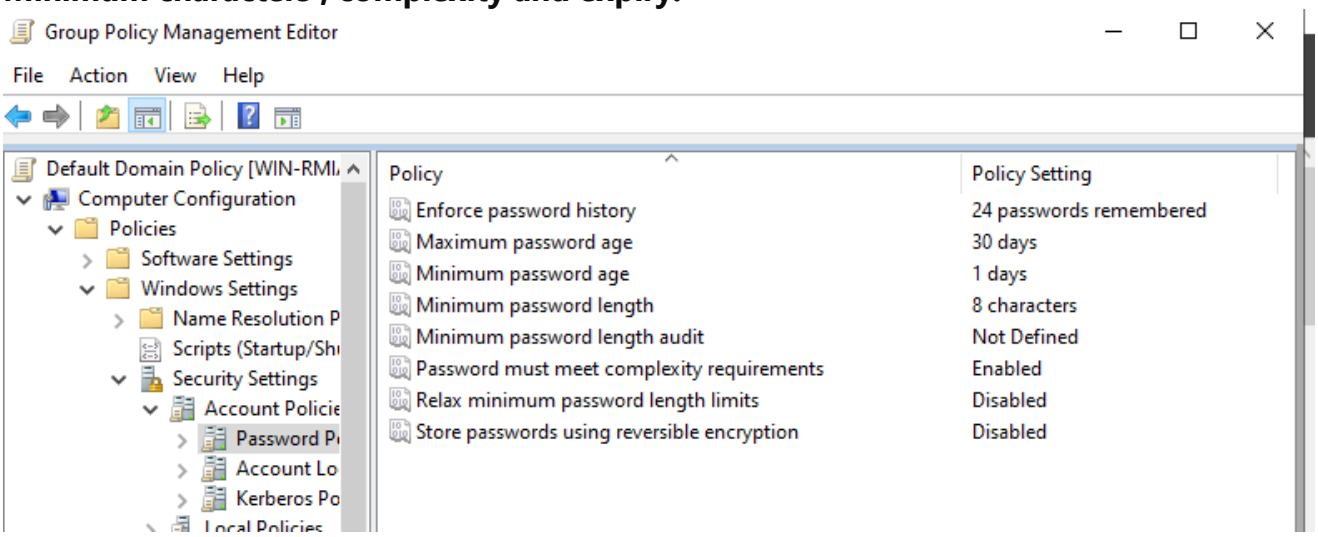
Group Policy Management

Set:

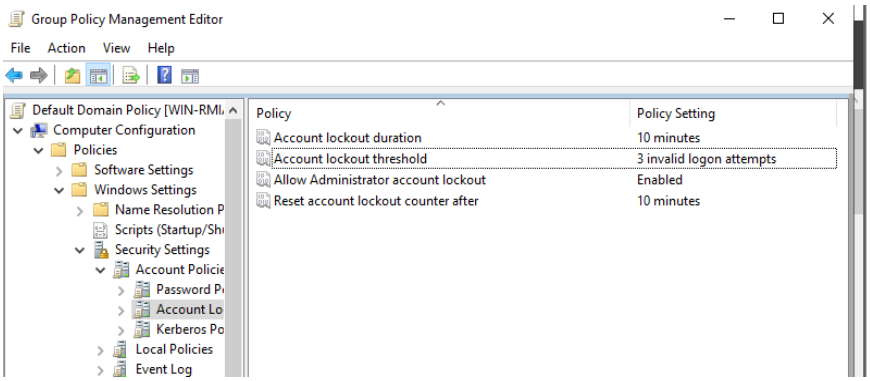
- ✓ Minimum 8 characters
- ✓ Complexity enabled
- ✓ Password expiry = 30 days
- ✓ Account lockout after 3 attempts

Password policy:

Minimum characters , complexity and expiry:



Account lockout:



Step 3: DHCP Configuration

On Ubuntu:

```
sudo apt install isc-dhcp-server
```

Configured DHCP range:

```
192.168.10.100 - 192.168.10.200
```

```
hambata@hambata-VMware-Virtual-Platform:~$ sudo apt install isc-dhcp-serv
er
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer req
uired:
  libconfig-inifiles-perl libdbd-mysql-perl libmariadb3
  libmysqlclient21 libterm-readkey-perl mariadb-common
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  isc-dhcp-common
```

Step 4: Security Measures Implemented

We applied the following security controls:

- Firewall enabled
- Closed unused ports
- Antivirus installed
- Anti-spam enabled
- Strong password policy
- Static IPs for servers
- Domain authentication
- Restricted Samba permissions

📸 **Insert Screenshot 17: Security settings**

4. Lab Challenges/Problems Encountered

Some problems I experienced were:

- Network issues between machines
- Trouble joining the domain at first
- Firewall blocking webmail
- Software's taking too long to download
- Network communication issues between VMs
- Difficulty joining the client to the domain
- Samba permission errors
- Firewall blocking webmail and shared folders

These were solved by reviewing the IP settings, permissions, and firewall rules.

5. Conclusion

In this lab, we successfully implemented different enterprise services in a virtual environment. We were able to configure the e-mail system, Active Directory, shared folders using Samba and NFS, DHCP, and password policies. We also applied different security measures to protect the systems and users.

Through this practical work, we improved our understanding of how enterprise systems are deployed and managed in real organizations. We also learned how Windows and Linux services can work together securely in a networked environment.

We learned how to design and implement a secure and reliable e-mail communication system for an enterprise environment. We were able to set up the virtual machines, configure the Windows Server as a Domain Controller, and join the client machine to the domain successfully. We also installed and configured the Linux mail server using Postfix, Dovecot, and Roundcube, which allowed users to send, receive, and access emails through a web browser.

In addition, we applied important security measures such as antivirus software, spam filtering, and firewall configuration to protect the system from threats and unauthorized access. Through this lab, we gained practical experience in server configuration, domain management, network setup, and e-mail security. Overall, the

lab helped us understand how different systems work together in a real business environment and improved our practical skills in operating systems security and administration.

References

Oracle. (2026). *VirtualBox Documentation*. https://www.virtualbox.org

Microsoft. (2026). *Windows Server Documentation*. https://learn.microsoft.com

Ubuntu. (2026). *Ubuntu Server Guide*. https://ubuntu.com/server/docs

Roundcube. (2026). *Roundcube Webmail*. https://roundcube.net

Postfix. (2026). *Postfix Mail Server Documentation*. http://www.postfix.org